

High Precision Wood-Wool Industrial Crafting Table (HPWICT)

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Wood Wool is Sustainable



- **Happy Camp Community** wants to use wood wool as a material that possesses the following properties:

- **Sustainable**
- **Insulating**
- **Fire Resistant**
- **Affordable**
- **Biodegradable**



Fig. 1

Insulating board made from wood wool



Fig. 2

Wood wool strands

Current Technology Has Limitations

- **Excelsior Wood Wool** machines are capable of producing **high outputs** of wood wool but have many **drawbacks**.

- Take up a **large area**
- Require **high power** input
- **Costly**
- **Dangerous** (fast moving inertia driven blade)
- **Difficult to use** (controls are a "black box")



Fig. 3

Excelsior Wood Wool Machine: Uses a large fly wheel to move blade along linear path.

Control of Fiber Dimensions Through Blade Geometry

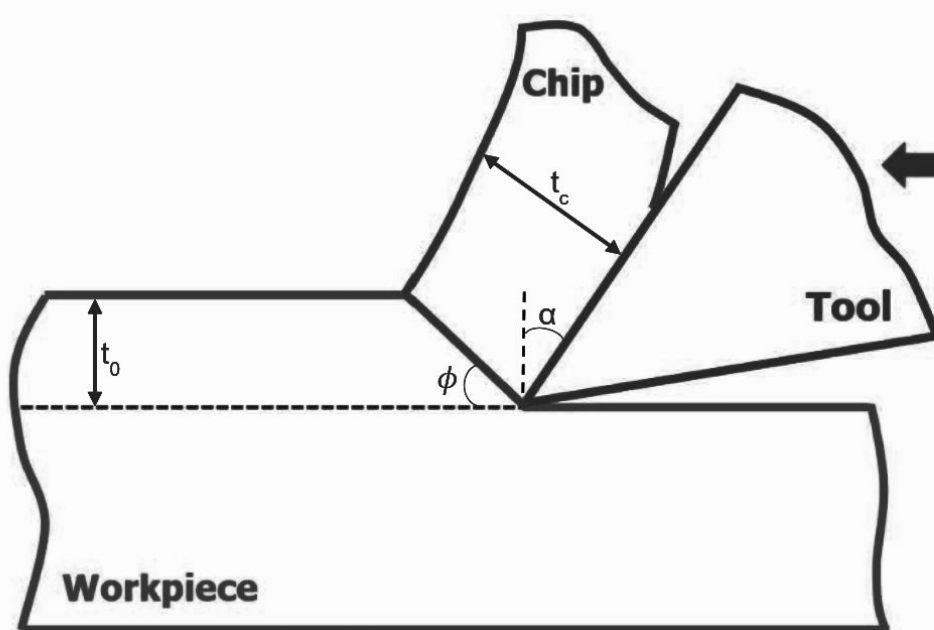


Fig. 4

Source: Lee, J. (2025). Topic 07: Forces in Cutting. Manufacturing Processes, LibreTexts.

Rake angle α	17	deg
Inclination angle λ	23.5	deg
Normal rake angle α_n	15.66	deg
Normal friction angle β_n	26.77	deg
Predicted shear angle ϕ	33.98	deg
Uncut chip thickness t_0	1.00	mm
Predicted chip thickness t_c	1.703	mm

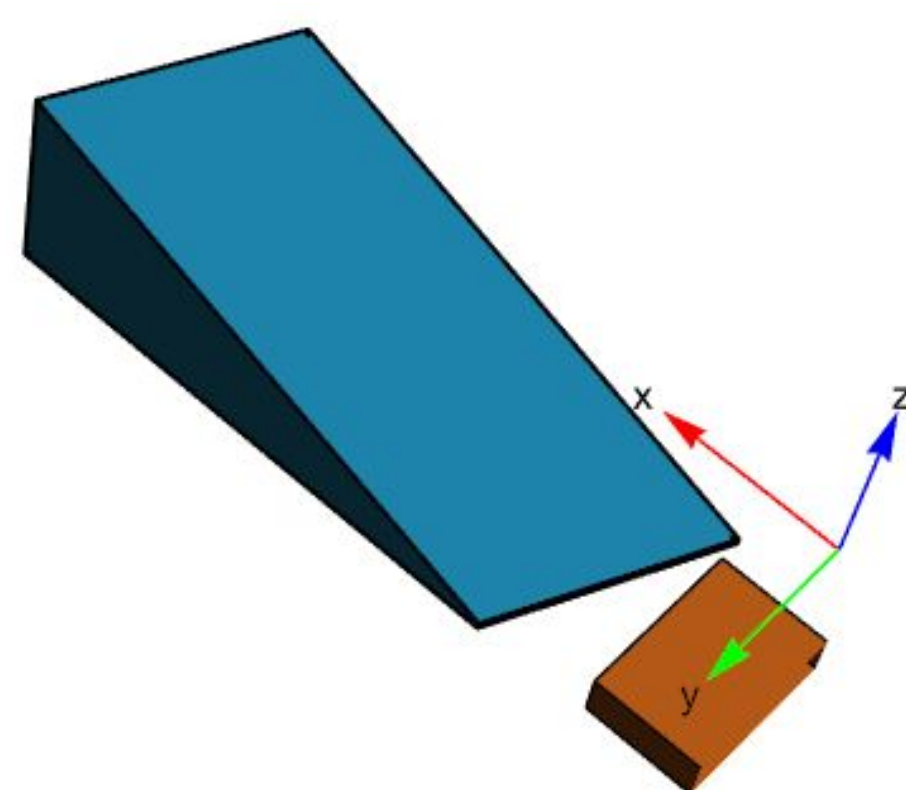


Fig. 5

Interactive Mathematica notebook where blade geometry is manipulated to change chip thickness.

Oblique Lee-Shaffer Relation

$$\phi = 45 + \alpha_n - \beta_n$$

Normal Rake Angle

$$\alpha_n = \tan^{-1}\left(\frac{\tan \alpha}{\cos \lambda}\right)$$

Normal Friction Angle

$$\beta_n = \tan^{-1}(\mu \cos \lambda)$$

Measuring Force and Chip Geometry



Fig. 6

Shimadzu UTM experimental setup.

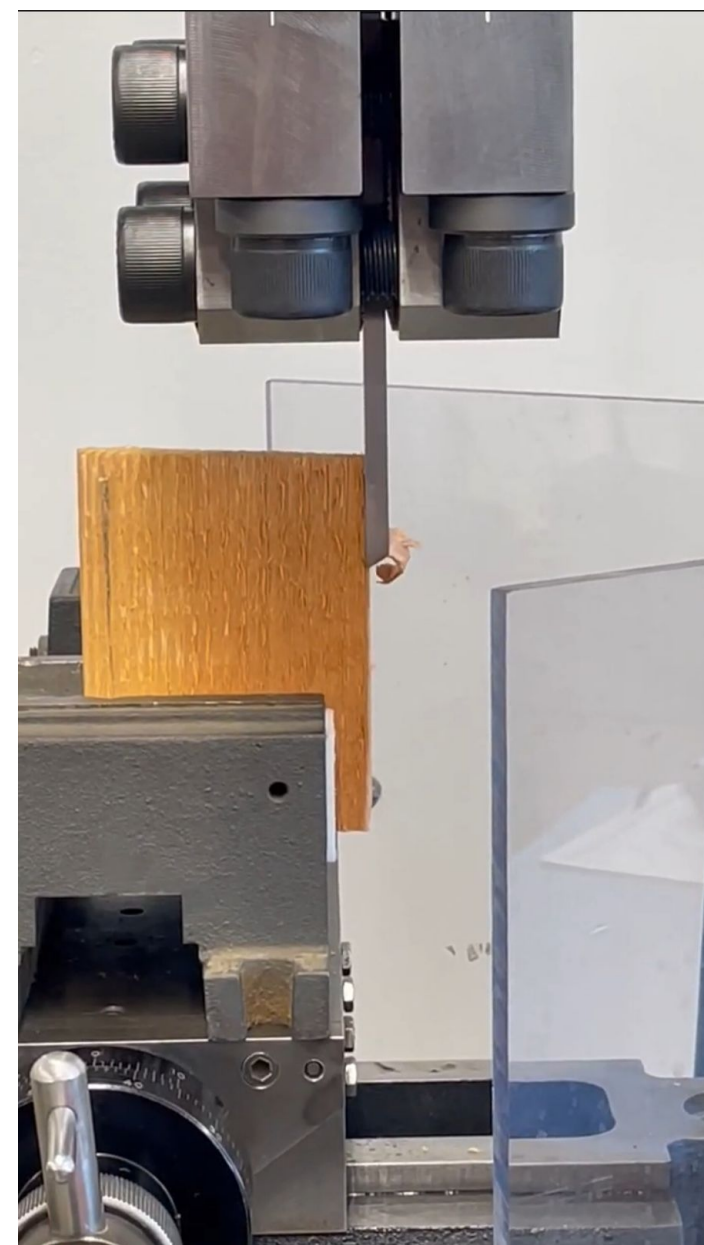


Fig. 7

Shimadzu UTM during the cutting experiments.

Douglas Fir

Cedar



Fig. 8

Resulting wood chips from a single experiment run.

- Two wood types, **Cedar** and **Douglas Fir**, were tested. The resulting chips were measured to evaluate blade geometry and determine which conditions produced **thin, consistent wood strands**.
- **Cutting Experiments** included two depths of cut for each wood type across three blade designs, resulting in **12 chip sets** for analysis.

Table Top HPWICT System

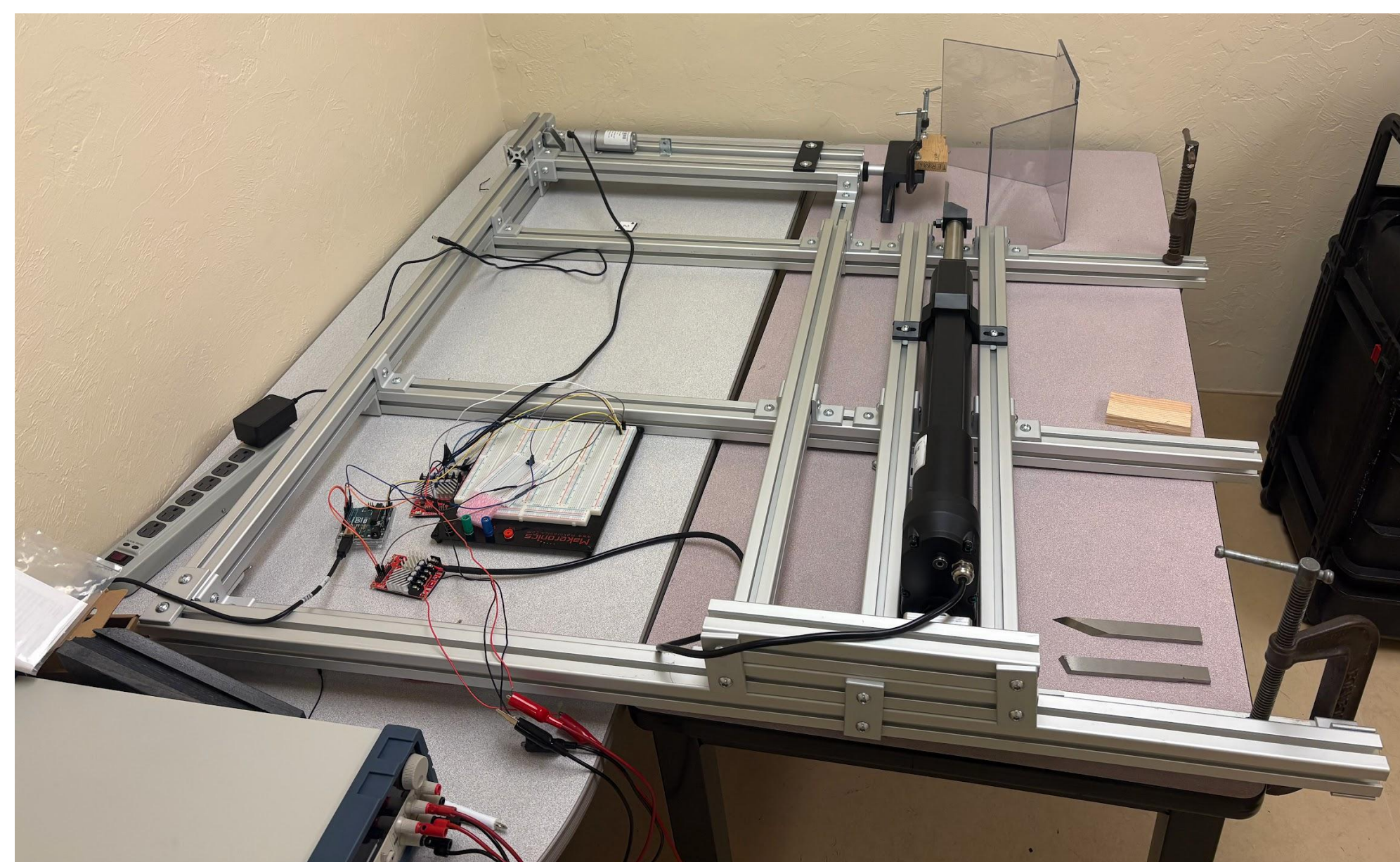


Fig. 9

Final Table Top assembly with controls components, cutting, and feedback linear actuators clamped down, and blade attached.

- **Control System:** The control system consists of:

- Two motor controllers
- Arduino R3 uno
- Power-supply
- Stop button
- Breadboard
- Wiring
- Feedback linear actuator
- Cutting linear actuator

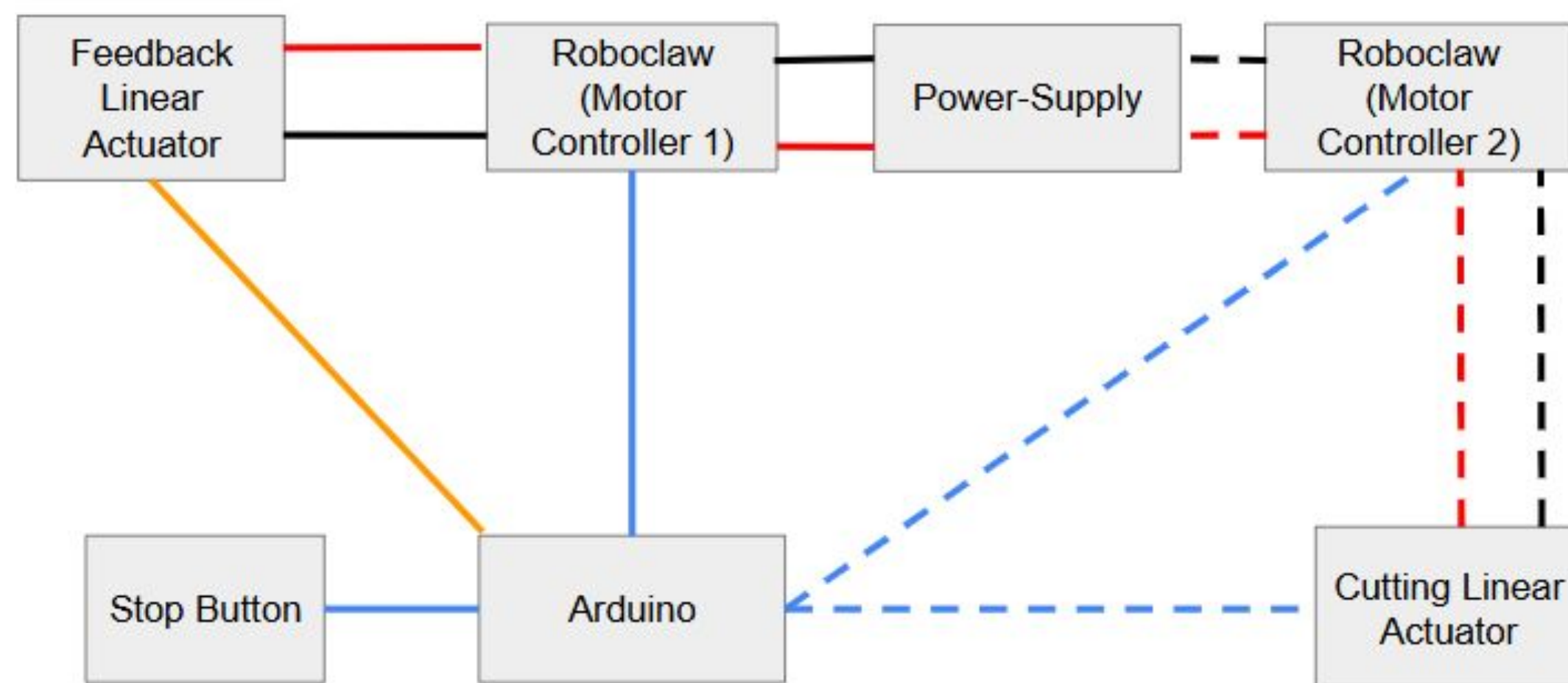


Fig. 10

The red and black solid lines represent positive (red) and negative (black). The blue solid & dash line represents digital signals. The orange solid line is the analog signal. And the red/black dash line represents motor control 1A/2A (red) and motor control 1B/2B (black).

Experimental Chip Geometry Matches Theory Prediction

Set	Blade	Depth of Cut [mm]	Wood Type
1	1	2	Cedar
2	1	1	Cedar
3	1	2	Douglas Fir
4	1	1	Douglas Fir
5	2	2	Douglas Fir
6	2	1	Douglas Fir
7	2	2	Cedar
8	2	1	Cedar
9	3	2	Cedar
10	3	1	Cedar
11	3	2	Douglas Fir
12	3	1	Douglas Fir

- **Figure 11** shows that the theoretical chip thickness predictions (red) overlap with the 95% uncertainty range of the experimental results, with a **p-value greater than 0.05**. This agreement indicates that chip dimensions **can be predicted** using fundamental cutting mechanics. The results also show that blade condition, wood type, and depth of cut affect chip consistency. Which provide guidance for improving wood wool quality and future machine design.

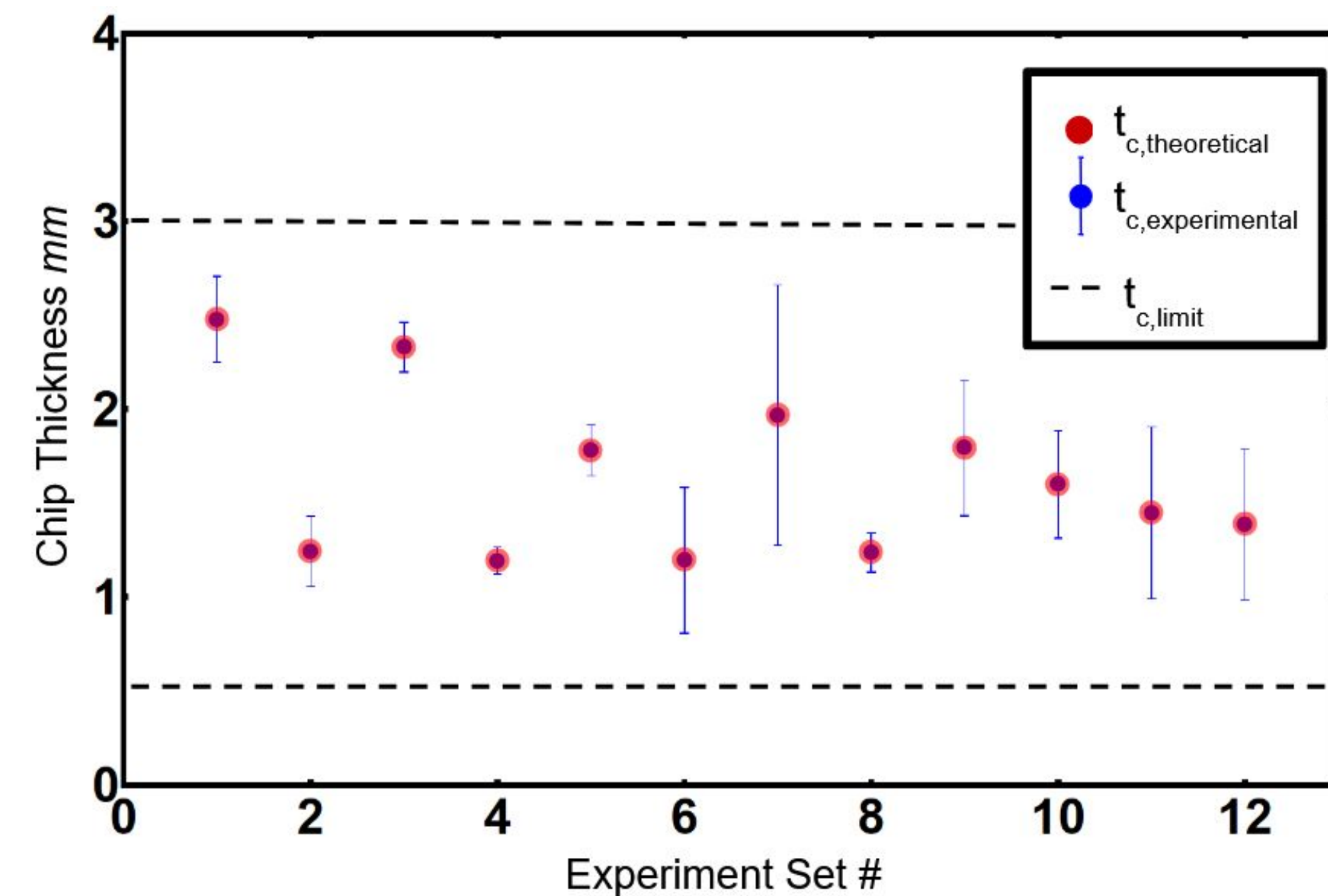


Fig. 11

Average chip thickness for each experimental blade test set, with red markers representing the theoretical chip thickness and blue error bars showing the experimental results and variation in measured thicknesses. The results highlight differences in chip consistency across the 12 experiments.

Conclusion

- ✓ Safe operation
- ✓ Control of wood wool dimensions through blade design
- ✓ Simple, robust platform for blade geometry testing
- ✓ Portable tabletop design
- ✓ Low-cost system



Fig. 12

CAD rendering of the tabletop wood wool cutting machine, designed with a T-slot frame for structural support and integrated actuators for blade motion and wood advancement.



Fig. 13

Image segmentation results showing isolated wood wool strands and fragments used for dimensional analysis, with a quarter as the scaling reference.

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Selected References

- Wood Handbook: Wood as an Engineering Material
 - <https://research.fs.usda.gov/fpl/wood-handbook>
- Utilization of geopolymer in wood wool insulation boards
 - <https://www.sciencedirect.com/science/article/pii/S0921344924001058>